Security Log Investigation

Analysis of a synthetic authentication log from three sources (VPN gateway, web portal, SSH bastion) covering 14 March 2026. I normalized the data, wrote seven detection rules, ran them, and rebuilt the incident timeline, keeping what the logs prove apart from what I am assuming.

> **Result:** two compromised accounts. **jdoe** through password spraying (02:10–02:52 UTC) and **admin** through brute force (03:05–03:22 UTC), each followed by security changes and large exports. One extra login (alice) needs follow-up.
> 149 raw lines, 147 normalized events (54 successes, 93 failures), 2 quarantined, 11 distinct IPs, 21 alerts

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1. Normalization

The three sources disagree on almost every field. Before any detection, I converted them to one schema.

Field	Raw variants	Normalized to
Timestamp	ISO 8601 with +03:00 (VPN), Unix epoch (portal), `Mar 14 02:12:44` with no year or zone (bastion)	ISO 8601 in UTC. For the bastion *assume* year 2026 and UTC.
IP	`src=`, `ip=`, `from ... port N`	One `ip` field, validated with Pytho `ddress`. Invalid values are quarant
Account	`Alice`, `ALICE`, `alice`, `invalid user root`	Lowercase, trimmed, then checke against the known-accounts list.
Action	`action=login`, `evt=file_export`, "Accepted/Failed password"	`login`, `mfa_disable`, `role_change`, `_user`, `file_export`
Result	`SUCCESS`, `ok`, `Accepted` / `FAILURE`, `fail`, `Failed`	`success` or `failure`

Raw lines, one per source

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```
2026-03-14T05:10:00+03:00 vpn-gw auth: user=Alice src=203.0.113.45 action=login result=FAILURE
ts=1773455770 app=portal usr=jdoe ip=203.0.113.45 evt=login outcome=ok
Mar 14 01:00:03 bastion sshd[3592]: Accepted password for svc-backup from 192.0.2.50 port 56651
```

Normalized sample (key events)

Line	Time (UTC)	Source	User	IP	Action	Result	Extra
2	2026-03-14 02:10:00	vpn-gw	alice	203.0.113.45	login	failure	
3	2026-03-14 02:10:37	vpn-gw	bob	203.0.113.45	login	failure	
33	2026-03-14 02:33:00	vpn-gw	jdoe	203.0.113.45	login	success	
34	2026-03-14 02:36:10	webapp	jdoe	203.0.113.45	login	success	
35	2026-03-14 02:41:20	webapp	jdoe	203.0.113.45	mfa_disable	success	
36	2026-03-14 02:45:05	webapp	jdoe	203.0.113.45	role_change	success	role=admin
37	2026-03-14 02:52:40	webapp	jdoe	203.0.113.45	file_export	success	bytes=18400
38	2026-03-14 03:05:00	bastion	admin	198.51.100.77	login	failure	
39	2026-03-14 03:05:06	bastion	admin	198.51.100.77	login	failure	
94	2026-03-14 03:11:20	bastion	admin	198.51.100.77	login	success	
96	2026-03-14 03:16:30	webapp	admin	198.51.100.77	create_user	success	target=suppo helper

Quarantined lines

Two lines could not be trusted and were kept out of the analysis instead of being guessed at.

Line	Content	Reason
41	`ts=1773450000 app=portal usr=carol ip=999.12.1.4 evt=login outcome=ok`	invalid value ('999.12.1.4' does not ap be an IPv4 or IPv6 address)
121	`Mar 14 05:59:01 bastion kernel: [UFW BLOCK] IN=eth0 SRC=203.0.113.9 DST=10.0.0.5 PROTO=TCP`	unrecognized format

2. Detection rules

Each rule has an explicit threshold so it can be tuned and explained. Alerts fired on this log are in the last column.

ID	Name	Logic	Why it works
R1	Password spraying	One source IP fails against ≥10 distinct accounts within 30 minutes, with ≤3 attempts per account.	Low attempts per account avoid lockouts, so counting accounts per IP is the signal.
R2	Brute force	≥10 failed logins for one account from one IP within 10 minutes.	Many guesses against one target. Spraying never triggers it.
R3	Success after attack	A login succeeds (a) after ≥5 failures for the same account, IP and source in the prior 15 minutes, or (b) from an IP already flagged by R1 or R2.	Turns noise into a real incident: the attacker got in.
R4a	New source IP	Successful login from an IP outside the user's usual set (office and own home IP).	Baseline deviation.
R4b	Off-hours login	Human account logs in outside 04:00–15:00 UTC (07:00–18:00 Doha). Service accounts allow-listed.	Attackers work in their own time zone.

R4c	Impossible travel	Two successful logins for one user from different countries within 60 minutes.	One person cannot be in both places.
R4d	Sensitive action, unusual IP	mfa_disable, role_change, create_user, or file_export ≥100 MB from an IP outside the user's usual set.	Post-compromise actions that matter most.

Alerts fired

Time (UTC)	Rule	Severity	Title	Detail	Lo lir
02:10:00	R1	High	Password spraying	31 failures against 25 accounts (16 do not exist), max 2 attempts per account	L2 (3
02:33:00	R3	Critical	Success from attacking IP	"jdoe" authenticated from 203.0.113.45, an IP already flagged for spraying or brute force	L3
02:33:00	R4a	Medium	Login from new source IP	"jdoe" logged in from 203.0.113.45 (Netherlands), not in their usual IPs	L3
02:33:00	R4b	Medium	Off-hours login	"jdoe" logged in at 02:33 UTC, outside 04:00-15:00	L3
02:36:10	R3	Critical	Success from attacking IP	"jdoe" authenticated from 203.0.113.45, an IP already flagged for spraying or brute force	L3
02:36:10	R4a	Medium	Login from new source IP	"jdoe" logged in from 203.0.113.45 (Netherlands), not in their usual IPs	L3
02:36:10	R4b	Medium	Off-hours login	"jdoe" logged in at 02:36 UTC, outside 04:00-15:00	L3
02:41:20	R4d	High	Sensitive action from unusual IP	"jdoe" did mfa_disable from 203.0.113.45	L3
02:45:05	R4d	High	Sensitive action from unusual IP	"jdoe" did role_change (role=admin) from 203.0.113.45	L3
02:52:40	R4d	High	Sensitive action	"jdoe" did file_export	L3

			from unusual IP	(bytes=184000000) from 203.0.113.45	
03:05:00	R2	High	Brute force on one account	55 failed logins for "admin" from one IP within 10 minutes	L3 (5
03:05:00	R3	Critical	Success after brute force	Login for "admin" succeeded right after 55 failures from the same IP	L3 (5
03:11:20	R4a	Medium	Login from new source IP	"admin" logged in from 198.51.100.77 (Romania), not in their usual IPs	L9
03:11:20	R4b	Medium	Off-hours login	"admin" logged in at 03:11 UTC, outside 04:00-15:00	L9
03:14:10	R3	Critical	Success from attacking IP	"admin" authenticated from 198.51.100.77, an IP already flagged for spraying or brute force	L9
03:14:10	R4a	Medium	Login from new source IP	"admin" logged in from 198.51.100.77 (Romania), not in their usual IPs	L9
03:14:10	R4b	Medium	Off-hours login	"admin" logged in at 03:14 UTC, outside 04:00-15:00	L9
03:16:30	R4d	High	Sensitive action from unusual IP	"admin" did create_user (target=support-helper) from 198.51.100.77	L9
03:22:00	R4d	High	Sensitive action from unusual IP	"admin" did file_export (bytes=1250000000) from 198.51.100.77	L9
09:20:14	R4c	Medium	Impossible travel	"alice": Qatar then Brazil within 21 minutes	L1 L1 lir
09:41:52	R4a	Medium	Login from new source IP	"alice" logged in from 203.0.113.131 (Brazil), not in their usual IPs	L1

Not alerted, on purpose: bob's four typos (below R3's threshold) and svc-backup's 01:00 login (allow-listed service account). No known false positives remain in this log.

3. Incident timeline

Times are UTC. The "what the log shows" column holds facts only; interpretation sits in its own column.

Time	What the log shows (evidence)	Lines	Interpretation (assumption)	Confide
01:00:03	svc-backup logs in over SSH from 192.0.2.50	L1	Scheduled nightly backup. Off-hours, but a known service account on its usual IP, so excluded from R4b.	Medium (assume schedule
02:10:00–02:33:00	31 failed VPN logins from 203.0.113.45 against 25 accounts, never more than 2 per account. 16 of the accounts do not exist.	L2–L32	Password spraying: a short list of common passwords tried across many accounts to stay under per-account lockouts.	High
02:33:00	jdoe logs in to the VPN successfully from 203.0.113.45, the same IP that just failed against 24 other accounts.	L33	jdoe's password was one of the sprayed passwords (weak or reused). The password itself is not visible.	High
02:36:10	jdoe logs in to the web portal from the same IP.	L34	Attacker uses the VPN foothold to reach internal applications.	High
02:41:20	jdoe disables MFA.	L35	Attacker removes the second factor to keep access. The portal accepted this from a never-seen IP.	High
02:45:05	jdoe's role is changed to admin.	L36	Privilege escalation. The log does not show which account performed the change beyond jdoe's session.	Medium
02:52:40	jdoe exports 184 MB.	L37	Possible data theft. The log proves the export ran, not that the file left the network.	Medium

03:05:00–03:10:51	55 failed SSH logins for admin from 198.51.100.77 in about 6 minutes.	L38–L93	Automated brute force. Different IP and technique from the spray, so a link between the two is not proven.	High
03:11:20	admin logs in over SSH from the same IP.	L94	Admin password guessed. Total compromise of the highest-privilege account.	High
03:14:10	admin logs in to the web portal from the same IP.	L95	Attacker moves from the bastion to the application.	High
03:16:30	admin creates a new user, support-helper.	L96	Likely a persistence account. The name looks designed to blend in.	Medium
03:22:00	admin exports 1,250 MB.	L97	Possible bulk data theft, larger than the earlier export.	Medium
02:10:37–06:15:02	bob fails 4 times, then succeeds, from his usual home IP.	L3–L109	Typo. Only 4 failures, below the 5-failure threshold, so no alert. A good example of tuning against false positives.	High (be
09:20:14–09:41:52	alice succeeds from 192.0.2.10 (Qatar), then from 203.0.113.131 (Brazil) 21 minutes later.	L122–L123	Either a VPN/proxy or stolen credentials. No failed logins, MFA data or other signals to decide. Needs follow-up.	Low

4. Evidence vs assumptions

Evidence (in the logs)

- 31 failed logins from 203.0.113.45 across 25 accounts in 23 minutes, 2 attempts at most each (L2–L32).
- jdoe succeeded from that same IP,

Assumptions (not proven)

- Bastion timestamps are UTC and from 2026. The log has no year or zone.
- The two attacking IPs are separate

then disabled MFA, became admin and exported 184 MB (L33–L37).

- 55 failed admin SSH logins from 198.51.100.77 in about 6 minutes, then a success (L38–L93).
- That IP then logged in to the portal, created `support-helper` and exported 1.25 GB (L94–L97).
- alice logged in from two countries 21 minutes apart (L122–L123).

actors. Nothing links them, and the same operator is possible.

- jdoe's password was weak or reused. Only the success is visible.
- Country data comes from a synthetic lookup table.
- Exports mean data theft. The log shows the action, not the download.
- `support-helper` is a backdoor account.
- svc-backup's 01:00 login is a scheduled job.
- alice's Brazil login is a VPN or compromise. Both fit.

Unknown (needs more data)

- What the new admin role and `support-helper` were used for afterwards.
- Whether the exported files left the network (proxy, DLP, endpoint logs).
- Whether MFA was enforced at jdoe's login, and how it could be disabled from a new IP.
- How the attacker learned the account names (the spray included real ones).
- Whether other accounts were quietly accessed with no log entry.

5. Recommended response

Contain now block 203.0.113.45 and 198.51.100.77, kill active sessions, reset jdoe and

- **Contain now:** block 203.0.113.45 and 198.51.100.77, kill active sessions, reset jdoe and admin passwords.
- **Undo changes:** re-enable MFA for jdoe, revert the role change, disable and review `support-helper`.
- **Investigate:** check what both exports contained, review actions by the new and elevated accounts, and look for the same IPs elsewhere.
- **Prevent:** enforce MFA on VPN and SSH, add per-account lockout, require re-authentication for MFA changes, alert on R1–R4d, and ban weak passwords.
- **Follow up:** ask alice whether she travelled or used a proxy.

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6. Short explanation

I first made the three log formats comparable: one timezone, one IP field, lowercase accounts, and two result values. Then I wrote rules around attacker behavior. Spraying is many accounts with few tries from one IP, brute force is many tries on one account, and unusual access is a login or action that does not match how the account is normally used. Running them gave a clear picture: a spray that cracked jdoe, then a separate brute force that cracked admin, each followed by risky actions.

I kept evidence and assumptions apart because logs show actions, not intent. For example, the export lines prove an export happened, but not that data left the company. The two thresholds that mattered most were R3's five-failure limit, which lets bob's typos through, and the off-hours window, which needs a service-account allow-list.

Appendix

▶ **Raw log (149 raw lines, 147 normalized events (54 successes, 93 failures), 2 quarantined, 11 distinct IPs, 21 alerts)**

▶ **Normalized events (CSV)**

▶ **analyze.py (normalization and detection rules)**

▶ gen.py (how the synthetic log was generated)